

# Solved Examples — Prime and Non-prime Attributes

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## Examples

**Example 1: Relation:**  $R(A, B, C, D, E, F)$

**FDs:**  $\{ AB \rightarrow C, C \rightarrow D, D \rightarrow E, F \rightarrow B, E \rightarrow F \}$

**Detailed Reasoning & Candidate Keys:**

- From the chain of FDs, starting at  $C$ :

$$C \rightarrow D, \quad D \rightarrow E, \quad E \rightarrow F, \quad F \rightarrow B$$

This implies:

$$C^+ = \{C, D, E, F, B\}$$

- Note that attribute  $A$  is not on the RHS of any FD, so it cannot be derived from other attributes.
- Therefore, any candidate key must include  $A$ .
- Compute closure of  $AB$ :

$$\begin{aligned} (AB)^+ &= \{A, B\} \xrightarrow{AB \rightarrow C} \{A, B, C\} \xrightarrow{C \rightarrow D} \{A, B, C, D\} \\ &\xrightarrow{D \rightarrow E} \{A, B, C, D, E\} \xrightarrow{E \rightarrow F} \{A, B, C, D, E, F\} \end{aligned}$$

All attributes are covered, so  $AB$  is a candidate key.

- Check closure of other combinations with  $A$ :

$$\begin{aligned} (AC)^+ &= \{A, C\} \xrightarrow{C \rightarrow D} \{A, C, D\} \xrightarrow{D \rightarrow E} \{A, C, D, E\} \\ &\xrightarrow{E \rightarrow F} \{A, C, D, E, F\} \xrightarrow{F \rightarrow B} \{A, B, C, D, E, F\} \end{aligned}$$

Similarly,  $(AD)^+, (AE)^+, (AF)^+$  all produce all attributes by similar logic.

**Candidate keys:**  $\{AB, AC, AD, AE, AF\}$

**Prime attributes:** All attributes appear in some candidate key:  $A, B, C, D, E, F$

**Non-prime attributes:** None

**Example 2: Relation:  $R(A, B, C, D, E, F)$** **FDs:**  $\{AB \rightarrow C, C \rightarrow DE, E \rightarrow F, C \rightarrow B\}$ **Detailed Reasoning & Candidate Keys:**

- From  $C$ , the closure is:

$$C \rightarrow D, E, B, F \Rightarrow C^+ = \{B, C, D, E, F\}$$

- Attribute  $A$  is not on the RHS of any FD, so every candidate key must include  $A$ .
- Compute closure of  $AC$ :

$$(AC)^+ = \{A, C\} \cup C^+ = \{A, B, C, D, E, F\}$$

- Similarly,

$$(AB)^+ = \{A, B\} \xrightarrow{AB \rightarrow C} \{A, B, C\} \xrightarrow{C \rightarrow DE, B, F} \{A, B, C, D, E, F\}$$

- Both  $AC$  and  $AB$  cover all attributes and hence are candidate keys.

**Candidate keys:**  $\{AB, AC\}$ **Prime attributes:**  $A, B, C$ **Non-prime attributes:**  $D, E, F$ **Example 3: Relation:  $R(A, B, C, D, E, F, G, H, I, J)$** **FDs:**  $\{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$ **Detailed Reasoning & Candidate Keys:**

- Compute  $A^+$ :

$$A^+ = \{A\} \xrightarrow{A \rightarrow DE} \{A, D, E\} \xrightarrow{D \rightarrow IJ} \{A, D, E, I, J\}$$

- Compute  $B^+$ :

$$B^+ = \{B\} \xrightarrow{B \rightarrow F} \{B, F\} \xrightarrow{F \rightarrow GH} \{B, F, G, H\}$$

- Combine  $A$  and  $B$ :

$$\begin{aligned} (AB)^+ &= A^+ \cup B^+ \cup \{C\} \quad \text{because } AB \rightarrow C \\ &= \{A, B, C, D, E, F, G, H, I, J\} \end{aligned}$$

- Hence,  $AB$  is a candidate key.
- Neither  $A$  nor  $B$  alone covers all attributes, so no single attribute key exists.

**Candidate key:**  $\{AB\}$ **Prime attributes:**  $A, B$ **Non-prime attributes:**  $C, D, E, F, G, H, I, J$

**Example 4: Relation:  $R(A, B, D, L, P, T)$**

**FDs:**  $\{B \rightarrow PT, A \rightarrow D, T \rightarrow L\}$

**Detailed Reasoning & Candidate Keys:**

- Compute  $B^+$ :

$$B^+ = \{B\} \xrightarrow{B \rightarrow PT} \{B, P, T\} \xrightarrow{T \rightarrow L} \{B, P, T, L\}$$

- Compute  $A^+$ :

$$A^+ = \{A\} \xrightarrow{A \rightarrow D} \{A, D\}$$

- Combine  $A$  and  $B$ :

$$(AB)^+ = A^+ \cup B^+ = \{A, B, D, P, T, L\}$$

- All attributes are covered, so  $AB$  is a candidate key.
- $A$  or  $B$  alone do not cover all attributes.

**Candidate key:**  $\{AB\}$

**Prime attributes:**  $A, B$

**Non-prime attributes:**  $D, P, T, L$

**Example 5: Relation:  $R(A, B, C, D, E, F, G, H)$**

**FDs:**  $\{E \rightarrow G, AB \rightarrow C, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, BC \rightarrow A\}$

**Step 1: Identify attributes that must be in every candidate key**

- Attributes  $F$  and  $H$  never appear on the right-hand side (RHS) of any FD, so they cannot be functionally determined by others.
- Therefore,  $F$  and  $H$  must be included in every candidate key.

**Step 2: Analyze cyclic dependencies among  $A, B, C$**

- The FDs  $AB \rightarrow C$ ,  $AC \rightarrow B$ , and  $BC \rightarrow A$  form a cycle among  $A, B, C$ .
- This means any two of  $A, B, C$  determine the third.

**Step 3: Check candidate keys with  $F$  and  $H$**

- Consider  $\{A, B, F, H\}$ :

$$(A, B, F, H)^+ \supseteq \{A, B, F, H\}$$

Using  $AB \rightarrow C$ , add  $C$ :

$$\{A, B, C, F, H\}$$

Using  $B \rightarrow D$ , add  $D$ :

$$\{A, B, C, D, F, H\}$$

Using  $AD \rightarrow E$  (since  $A$  and  $D$  are in closure), add  $E$ :

$$\{A, B, C, D, E, F, H\}$$

Using  $E \rightarrow G$ , add  $G$ :

$$\{A, B, C, D, E, F, G, H\}$$

All attributes covered, so  $\{A, B, F, H\}$  is a candidate key.

- Similarly,  $\{A, C, F, H\}$  and  $\{B, C, F, H\}$  are candidate keys because any pair from  $A, B, C$  determines the third.

**Step 4: Determine prime and non-prime attributes**

- **Candidate keys:**  $\{A, B, F, H\}, \{A, C, F, H\}, \{B, C, F, H\}$
- **Prime attributes:** All attributes appearing in any candidate key:

$$A, B, C, F, H$$

- **Non-prime attributes:** Attributes that do not appear in any candidate key:

$$D, E, G$$

**Example 6: Relation:  $R(A, B, C, D, E)$** **FDs:**  $\{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$ **Step 1: Compute closures of candidate attribute sets**

- Calculate  $E^+$ :

$$E^+ = \{E\}$$

Using  $E \rightarrow A$ , add  $A$ :

$$E^+ = \{E, A\}$$

Using  $A \rightarrow BC$ , add  $B, C$ :

$$E^+ = \{E, A, B, C\}$$

Using  $B \rightarrow D$ , add  $D$ :

$$E^+ = \{E, A, B, C, D\}$$

All attributes are included, so  $E$  is a candidate key.

- Calculate  $A^+$ :

$$A^+ = \{A\}$$

Using  $A \rightarrow BC$ , add  $B, C$ :

$$A^+ = \{A, B, C\}$$

Using  $B \rightarrow D$ , add  $D$ :

$$A^+ = \{A, B, C, D\}$$

Using  $CD \rightarrow E$ , since  $C$  and  $D$  are in closure, add  $E$ :

$$A^+ = \{A, B, C, D, E\}$$

All attributes are included, so  $A$  is a candidate key.

- Calculate  $CD^+$ :

$$CD^+ = \{C, D\}$$

Using  $CD \rightarrow E$ , add  $E$ :

$$CD^+ = \{C, D, E\}$$

Using  $E \rightarrow A$ , add  $A$ :

$$CD^+ = \{A, C, D, E\}$$

Using  $A \rightarrow BC$ , add  $B$ :

$$CD^+ = \{A, B, C, D, E\}$$

All attributes are included, so  $CD$  is a candidate key.**Step 2: Identify prime and non-prime attributes**

- **Candidate keys:**  $\{E, A, CD\}$
- **Prime attributes:** Attributes that appear in at least one candidate key:

$$A, C, D, E$$

- **Non-prime attributes:** Attributes that do not appear in any candidate key:

$$B$$

**Example 7: Relation:  $R(A, B, C, D, E, H)$** **FDs:**  $\{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A\}$ **Step 1: Identify attributes not appearing on RHS**

Observe that attribute  $H$  does not appear on the right-hand side of any FD. This means:

- $H$  cannot be functionally determined by any other attributes.
- Therefore,  $H$  **must** be part of every candidate key.

**Step 2: Explore dependencies and closures**

- Given the FDs, note the cyclic dependencies among  $A, B, D$ :

$$D \rightarrow A, \quad A \rightarrow B, \quad BC \rightarrow D$$

- Calculate closures of key attribute sets including  $H$  to find candidate keys:
- $(A, E, H)^+$ : Start with  $\{A, E, H\}$ .
  - From  $A \rightarrow B$ , add  $B \rightarrow \{A, B, E, H\}$ .
  - From  $E \rightarrow C$ , add  $C \rightarrow \{A, B, C, E, H\}$ .
  - From  $BC \rightarrow D$ , since  $B, C$  are in closure, add  $D \rightarrow \{A, B, C, D, E, H\}$ .
  - From  $D \rightarrow A$ ,  $A$  already included.

Closure = all attributes  $\Rightarrow (A, E, H)$  is a candidate key.

- $(B, E, H)^+$ : Start with  $\{B, E, H\}$ .
  - From  $E \rightarrow C$ , add  $C \rightarrow \{B, C, E, H\}$ .
  - From  $BC \rightarrow D$ , add  $D \rightarrow \{B, C, D, E, H\}$ .
  - From  $D \rightarrow A$ , add  $A \rightarrow \{A, B, C, D, E, H\}$ .
  - From  $A \rightarrow B$ ,  $B$  already included.

Closure = all attributes  $\Rightarrow (B, E, H)$  is a candidate key.

- $(D, E, H)^+$ : Start with  $\{D, E, H\}$ .
  - From  $D \rightarrow A$ , add  $A \rightarrow \{A, D, E, H\}$ .
  - From  $A \rightarrow B$ , add  $B \rightarrow \{A, B, D, E, H\}$ .
  - From  $E \rightarrow C$ , add  $C \rightarrow \{A, B, C, D, E, H\}$ .
  - From  $BC \rightarrow D$ , already included.

Closure = all attributes  $\Rightarrow (D, E, H)$  is a candidate key.

**Step 3: Summary of candidate keys and attribute classification**

- Candidate keys are:

$$\{A, E, H\}, \quad \{B, E, H\}, \quad \{D, E, H\}$$

- **Prime attributes:** Those that appear in at least one candidate key:

$$A, B, D, E, H$$

- **Non-prime attributes:** Those not appearing in any candidate key:

$$C$$

### Notes & Tips

- Attributes not appearing on the RHS of any FD must be in every candidate key.
- Start closures with smaller attribute sets, especially those frequently appearing on the LHS.
- Cyclic FD sets often imply candidate keys formed by pairs or triplets.
- Prime attributes appear in at least one candidate key; others are non-prime.